Avionics Navigation Systems (ECE4635/ESE482)

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Sheet One

Theoritical Questions:

1. **What is the main purpose of a navigation system?**

To determine the aircraft's position and motion and support movement toward the desired destination.

2. **What is the difference between navigation and guidance?**

Navigation determines where the aircraft is, while guidance determines where it should go.

3. **What is the role of the control system?**

To make the aircraft follow the desired path generated by the guidance system.

4. **Why is integrated navigation useful?**

Because it combines information from multiple systems and can provide a more accurate and reliable navigation solution.

5. **Why can an inertial navigation system develop errors over time?**

Because small sensor errors accumulate during integration, producing increasing navigation error or drift.

6. **Give examples of self-contained navigation systems.**

A self-contained navigation system determines the aircraft's position and motion without relying on external signals or ground-based navigation aids.

Examples include:

- Inertial Navigation System (INS)
- Inertial Reference System (IRS)
- Dead-Reckoning Navigation System

The most important example is the INS, which uses gyroscopes and accelerometers to determine the aircraft's position, velocity, and attitude.

7. **Why can an INS accumulate error with time?**

   An INS calculates position by integrating measured acceleration over time. Small errors in the accelerometers and gyroscopes are also integrated, causing the navigation error to increase with time.

   For example: Acceleration error → Velocity error → Position error

   INS is highly accurate for short periods, but its position error tends to grow with time unless it is corrected by another navigation source.

8. **What is the advantage of integrated navigation?**

   Integrated navigation combines information from two or more navigation systems to obtain better overall performance. For example: INS + GPS

   The advantage is that the strengths of one system can compensate for the weaknesses of another.

   - INS: independent of external signals but accumulates errors.
   - GPS: provides accurate position but depends on satellite signals.

   Therefore, an integrated GPS/INS system can provide higher accuracy, better reliability, and continuous navigation.

9. **Why do we need different navigation systems during different phases of flight?**

   Different phases of flight have different navigation requirements in terms of accuracy, update rate, coverage, and reliability.

For example:

- Takeoff: accurate position and heading are important.
- En-route: long-range and continuous navigation are required.
- Approach: much higher positioning accuracy is needed.
- Landing: very high accuracy and reliability are required.

No single navigation system is necessarily optimal for every phase of flight. Different systems can be selected or integrated according to the requirements of each flight phase.

10. What are the three basic aircraft attitude angles?

The three basic aircraft attitude angles are:

1. **Roll** (ϕ)– rotation about the **longitudinal axis**.
2. **Pitch** (θ)– rotation about the **lateral/transverse axis**.
3. **Yaw** (ψ)– rotation about the **vertical axis**.